

Sea-ice sub-cycling in the FESOM2 GPU port: what pays, and what it costs

N. Koldunov, 6 August 2026 — summary of the mEVP communication study

Three ways to stop exchanging the halo 120 times per step

mEVP runs 120 sub-cycles per model time step and currently exchanges the ice-velocity halo after every one. Three ways to do it less often were implemented and measured:

delayed exchange — exchange every K -th sub-cycle and let the halo go stale. Fastest by far, *but it imprints the domain decomposition on the ice* (below). **Wide halo** (Danilov) — exchange every K -th sub-cycle but recompute a K -deep ghost ring in between, so nothing is stale: *exact*, bit for bit, and no imprint; smaller gain. **Divergence form** (Danilov) — carry $\nabla \cdot \sigma$ at nodes instead of σ at elements: exact to 10^{-15} , pays on CPU, costs on GPU.

What they buy — in both currencies

Percent of the *model step* is what a production run feels; percent of the *ice cost* (ice + ice dynamics + ice advection, compute *and* MPI wait) is what judges a sea-ice scheme. They differ 5–10 $\times$, and quoting only the first makes a good scheme look like nothing.

		GPU		CPU, at scale	
		ice cost	model step	ice cost	model step
delayed exchange	$K=4$	–37 to –58%	–8.5 to –14%	–5 to –25%	–0.3 to –3.6%
wide halo, exact	$K=8$	–12 to –19%	–2 to –7%	never wins	
divergence form alone	—	+2 to +17%	0 to +1.6%	–2 to –8.5%	–0.5 to 0%

The CPU columns grow with core count (–5% of the ice at 128 ranks, –25% at 2048); the GPU columns shrink with the mesh a GPU holds (–58% at 32 k nodes per GPU, –37% at 460 k). The divergence form is **not** a null result: on CPU at 512 ranks and above it takes 2–8.5% off the ice and is simply invisible in the model step. On GPU it costs, because there the sub-cycle is limited by the number of GPU operations issued rather than by arithmetic or memory traffic, so removing element arrays removes work that was not being paid for. Danilov’s own $\sim 20\%$ for the divergence form with its double halo matches our –19.1% **of the ice cost** for that combination (fArc, 16 GPU nodes).

The finding that decides the recommendation: the fast option imprints the partition

The delayed exchange reads halo values up to $K - 1$ sub-cycles old, and those nodes are exactly the one-cell rim of each sub-domain. The partition is static, so the same nodes are perturbed every step of the run. Measured on fArc, 16 GPUs, 60 days of daily output — mean $|\Delta \text{ ice thickness}|$ on the rim against the interior:

	rim	interior	ratio
two identical runs (control — no lever at all)	0.00105 m	0.00124 m	0.85
wide halo, exact, $K=8$	0.00104	0.00120	0.87
delayed exchange, $K=2$	0.00432	0.00249	1.70
delayed exchange, $K=4$	0.01178	0.00452	2.52

It is there on **day one** (ratio 3.1), persists for two months, scales with K , and traces the sub-domain boundaries visibly. The exact wide halo scores like the control, as it must, which also validates the diagnostic. **The delayed exchange therefore makes the answer depend on how the mesh was partitioned.** The ring profiles and the map are Figure 4 of the full note.

What the delayed exchange costs in simulated climate (1 year, CORE2, exactly reproducible)

Pattern correlation against the same run exchanging every sub-cycle, judged against the difference between two independent correct implementations of mEVP (our C port versus Fortran FESOM2). At $K=4$: ice concentration 0.999995 and thickness 0.999994 against a yardstick of 0.99999/0.99998; ice velocity 0.963/0.950 against 0.954/0.939. Every field stays inside the yardstick at $K \leq 8$, but only by 1.1–1.5 $\times$ on ice velocity. Hemispheric ice extent moves 0.23% of its seasonal amplitude and the ocean surface is unchanged to six digits. *CORE2 resolution only.*

Recommendation

For production, the exact wide halo. Bit-for-bit identical to the standard scheme, so no accuracy argument and no partition imprint, and worth 12–19% of the ice cost on GPU at $K=8$. It does not pay on CPU on any mesh.

The delayed exchange is not recommended as it stands, despite being worth three times as much: results stop being reproducible across core counts and the error is organised along artificial lines. If the speed is wanted, establish whether the imprint matters over a decade, or whether a hybrid (short delay, occasional exact refresh) removes it.

The divergence form is worth having on CPU at scale and is a cost on GPU — and it is what makes the wide halo cheapest to carry.